HYBRID AI-BASED DETECTION OF FILELESS MALWARE IN IoMT HEALTHCARE SYSTEMS

INDIVIDUAL THESIS

CHATHURANGA L A S

IT22108272

Component 01: DICOM Medical Imaging Network Anomaly Detection

Group ID: R26-CS-006

B.Sc. (Hons) Information Technology – Cyber Security

Sri Lanka Institute of Information Technology

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DECLARATION AND SUPERVISOR CERTIFICATION

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Student name: [CHATHURANGA L A S]

Student ID: [IT22108272]

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Supervisor certification. The above candidate carried out this research for the bachelor's degree thesis under my supervision.

Role and name	Signature	Date
Supervisor: Mr. Kanishka Yapa		
Co-supervisor: Mr. Dinith Primal		

ABSTRACT

Medical imaging systems exchange sensitive diagnostic studies through DICOM and PACS infrastructure, creating a need to detect abnormal communications while preserving image integrity. This individual research component of MediRecon investigates a hybrid workflow combining supervised network anomaly classification, controlled PACS traffic validation and SHA-256 integrity comparison. A CICIoMT-derived dataset and Random Forest approach support offline model development; a separate 28-feature live inference workflow collects network packets, aggregates flow descriptors and produces SOC-facing risk indications. Orthanc C-ECHO and C-STORE tests demonstrate basic imaging communication, while trusted and received DICOM files are compared through cryptographic hashes. Recorded live tests show successful feature extraction and abnormal traffic classifications, including a 308-packet window assigned a 74.50% abnormal-class probability. Such a probability is not equivalent to validated detection accuracy. The principal contribution is a modular and inspectable integration of network telemetry, imaging integrity and SOC decision support. Further held-out model evaluation, feature-schema validation and independent hospital-like testing are required before claims of operational effectiveness can be made.

Keywords: DICOM, IoMT, Random Forest, network anomaly detection, SHA-256

ACKNOWLEDGEMENTS

The author acknowledges the guidance of Mr. Kanishka Yapa and Mr. Dinith Primal, the support of the SLIIT academic staff, and the contributions of project team members who developed the complementary components of MediRecon. All substantive technical assistance should be reviewed and acknowledged before submission.

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
API	Application Programming Interface
C-ECHO	DICOM Verification Service
C-STORE	DICOM Storage Service
CSV	Comma-Separated Values
DICOM	Digital Imaging and Communications in Medicine
HTTP	Hypertext Transfer Protocol
HTTPS	Hypertext Transfer Protocol Secure
IoMT	Internet of Medical Things
IP	Internet Protocol
JSON	JavaScript Object Notation
ML	Machine Learning
PACS	Picture Archiving and Communication System
RF	Random Forest
SHA-256	Secure Hash Algorithm 256-bit
SOC	Security Operations Centre
TCP	Transmission Control Protocol
TShark	Terminal-based Wireshark packet analyser
XAI	Explainable Artificial Intelligence

1 INTRODUCTION

1.1 Overall project context

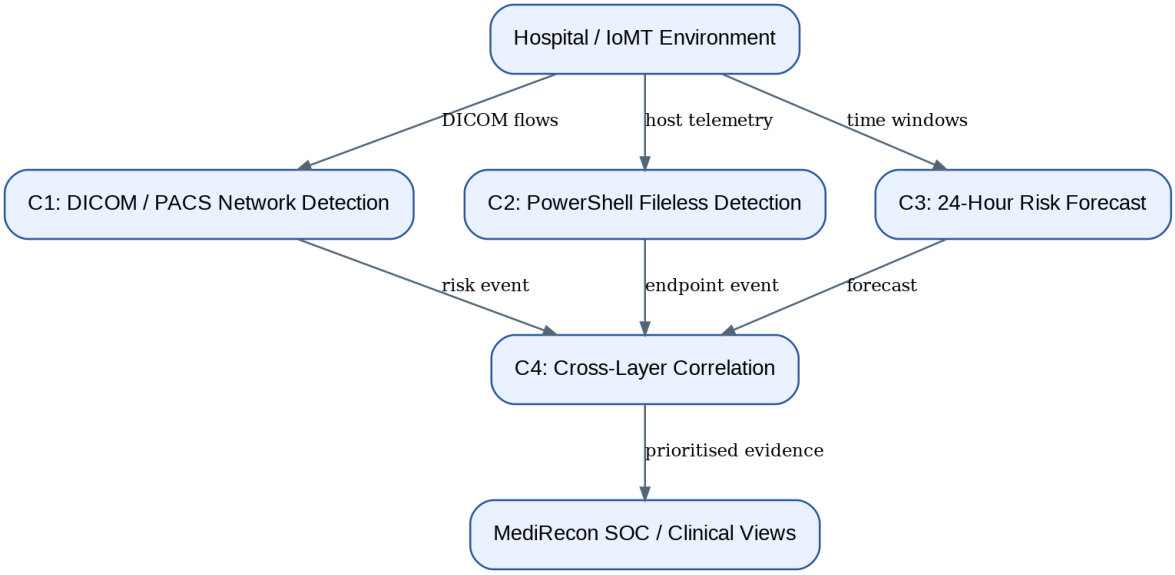


Figure 1.1: MediRecon overall research component architecture (author-created schematic).

MediRecon is a group research product addressing the detection and investigation of fileless-malware-related security risks in healthcare Internet of Medical Things (IoMT) environments. The overall project combines medical imaging network anomaly detection, PowerShell-based endpoint detection, next-24-hour threat forecasting and cross-layer correlation. This individual thesis focuses specifically on Component 01: DICOM Medical Imaging Network Anomaly Detection.

The individual component combines network-flow classification with medical image integrity checking. These are complementary but distinct detection functions: Random Forest processes numerical traffic descriptors, while SHA-256 compares trusted and received DICOM files. A suspicious flow does not by itself prove malware, and a matching file hash does not establish that the network was free of compromise.

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In a hospital context, a useful security decision requires an auditable distinction among measurement, classification and response. Packet-derived features describe observed network behaviour; a classifier estimates a learned class; severity and incident response are operational policies. A correct deployment should retain all three as separately inspectable fields and should support analyst review before high-impact containment.

1.2 Individual component context

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1.3 Problem statement

MediRecon is a group research product addressing the detection and investigation of fileless-malware-related security risks in healthcare Internet of Medical Things (IoMT) environments. The overall project combines medical imaging network anomaly detection, PowerShell-based endpoint detection, next-24-hour threat forecasting and cross-layer correlation. This individual thesis focuses specifically on Component 01: DICOM Medical Imaging Network Anomaly Detection.

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1.5 Aim and objectives

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1.8 Individual contribution

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1.9 Thesis structure

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2 LITERATURE REVIEW AND RELATED WORK

2.1 Review method and scope

DICOM specifies the communication and management of medical imaging information. Its standardisation enables interoperability among imaging modalities, archives and workstations; however, interoperable exchange does not remove the need for security monitoring [2], [3]. The CICIoMT2024 benchmark offers healthcare IoMT traffic associated with several attack categories and device configurations, but it is not a DICOM-specific fileless-malware ground-truth corpus [1].

Random Forest, introduced by Breiman, aggregates predictions from multiple decision trees and can model nonlinear relationships among tabular features [4]. Model suitability must be evaluated through reproducible experiments, not inferred from the algorithm name alone. The NIST AI RMF encourages measurement, governance and ongoing management of AI risks [5].

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2.2 Domain and theoretical foundations

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2.4 Existing systems products and techniques

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2.5 Gap synthesis and conceptual framework

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however, interoperable exchange does not remove the need for security monitoring [2], [3]. The CICIoMT2024 benchmark offers healthcare IoMT traffic associated with several attack categories and device configurations, but it is not a DICOM-specific fileless-malware ground-truth corpus [1].

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3 RESEARCH METHODOLOGY

3.1 Research design

The research uses a design-and-evaluate engineering approach. An offline supervised classification pipeline learns from a cleaned CICIoMT-derived traffic table; a separate live-capture pipeline aggregates observed packets into numerical features and supplies them to a saved classifier. The DICOM integrity sub-system computes SHA-256 digests of trusted and received studies. A practical PACS testbed supplies controlled traffic for system verification.

Reported development records describe approximately 600,001 source rows, a 44-feature training context, and a 28-feature live classifier. The exact retained-row count, class frequencies, training split, fitted hyperparameters and held-out evaluation scores must be extracted from the final notebook and model artefacts before the thesis is submitted. The two feature schemas must not be presented as interchangeable without evidence of a documented transformation or separately trained model.

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3.2 Data participants or study materials

The project distinguishes the offline training corpus from operational evidence. The offline CICIoMT-derived table was described during development as approximately 600,001 records with a 44-feature source representation and a binary label. This figure describes the source-table context and must not be treated as proof that every one of the 44 columns was used in the deployed model. The live network feature extractor creates 28 ordered numerical fields. The

deployed Random Forest must be validated against precisely those 28 fields; a 44-input estimator cannot legitimately receive a 28-input vector without a separately documented transformation or retraining.

Before final submission, the training notebook should supply the exact dataset filename, checksum, provenance, label distribution, cleaning operations, excluded identifiers, data split sizes, training feature order and exported model metadata. Label 0 denotes NORMAL and label 1 denotes ABNORMAL in the observed interface. Class balance must be computed from the exact training CSV rather than inferred from a separate CICIoMT release.

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3.3 Experimental or study procedure

Offline evaluation should begin with a deterministic random seed, an auditable train/test split and leakage checks. Feature engineering must be fit on training data only where learned transforms are involved. Hyperparameter search must be performed using a training/validation procedure; the held-out test partition must be reserved for final reporting. The source repository and model artefact should be archived together so that the inference feature contract is not reconstructed from memory.

Operational validation uses two connected Windows machines. The monitored host generates controlled HTTP/TCP traffic or DICOM C-ECHO/C-STORE requests; the security host observes the traffic using TShark/Npcap. The capture pipeline extracts packet-level measurements, aggregates them into a network window, constructs 28 feature values and executes Random Forest inference. Predicted class, class probability, severity and packet counts are persisted in SQLite and displayed through the SOC portal.

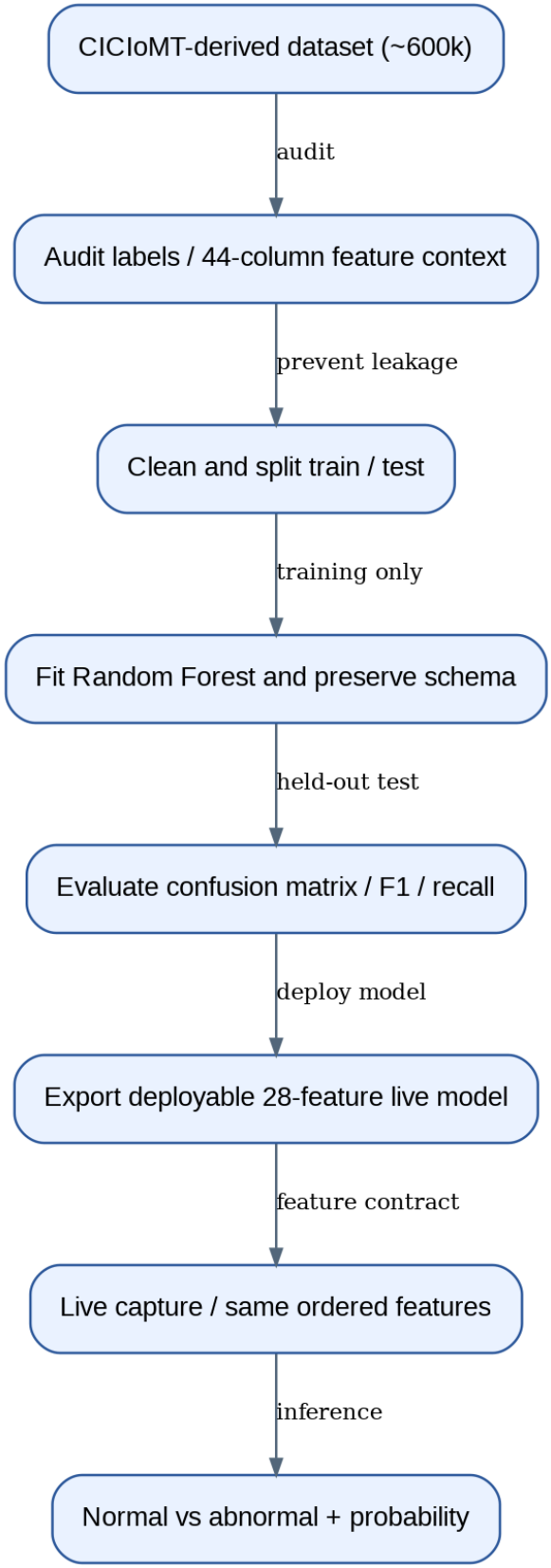


Figure 3.1: Dataset audit, Random Forest training and live feature-contract workflow.

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3.4 Measures and evaluation metrics

Table 3.1: Live network feature categories

Category	Examples
Flow statistics	Duration, Rate, Number
TCP flags	SYN, ACK, FIN, RST, PSH, ECE, CWR
Protocols	HTTP, HTTPS, DNS, TCP, UDP, ARP, ICMP, IGMP
Packet sizes	Min, Max, AVG, Std, Tot size, Variance

Accuracy alone is insufficient when abnormal events are rare. The thesis therefore specifies precision, recall, F1-score, true-positive rate, false-positive rate, confusion matrix and per-class support. Operational measurements should include capture-window duration, number of captured packets, feature-completeness rate, inference latency, database persistence success and event-delivery success. Where an evaluation image is unavailable, the corresponding numerical cell must remain marked NOT VERIFIED instead of displaying an assumed value.

An abnormal probability such as 0.885 is a model score for the current captured window, not evidence that a real-world attack has an 88.5% probability of occurring. The severity mapping is an application policy applied to model output. Severity thresholds should be documented and tested at exact boundary values.

The research uses a design-and-evaluate engineering approach. An offline supervised classification pipeline learns from a cleaned CICIoMT-derived traffic table; a separate live-capture pipeline aggregates observed packets into numerical features and supplies them to a saved classifier. The DICOM integrity sub-system computes SHA-256 digests of trusted and received studies. A practical PACS testbed supplies controlled traffic for system verification.

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policies. A correct deployment should retain all three as separately inspectable fields and should support analyst review before high-impact containment.

Verification checklist: preserve training/test split and random seed; report class frequencies; inspect feature ordering; capture fitted hyperparameters; include per-class precision, recall, F1 and confusion matrix; test calibration and false-positive rates; repeat results across multiple controlled captures.

3.5 Data analysis

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3.6 Ethics privacy security and responsible AI

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3.7 Reproducibility and research artefacts

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4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

The proposed architecture separates capture, feature engineering, model inference, image-integrity verification, persistence and presentation. A Windows-based Orthanc PACS service provides DICOM C-ECHO and C-STORE functionality; a second host initiates network traffic. Capture scripts record packet metadata and compute protocol ratios, TCP flag statistics, rates and packet-length statistics.

The SOC interface is intended for technical investigation, showing traffic classification, model-derived risk, source and destination context, evidence and alerts. The doctor-facing interface has a narrower clinical safety task: indicate whether a received image matches a registered trusted file. The integrated group platform consumes component events through an agreed schema rather than relying on shared assumptions about database tables.

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4.2 Component architecture within the complete solution

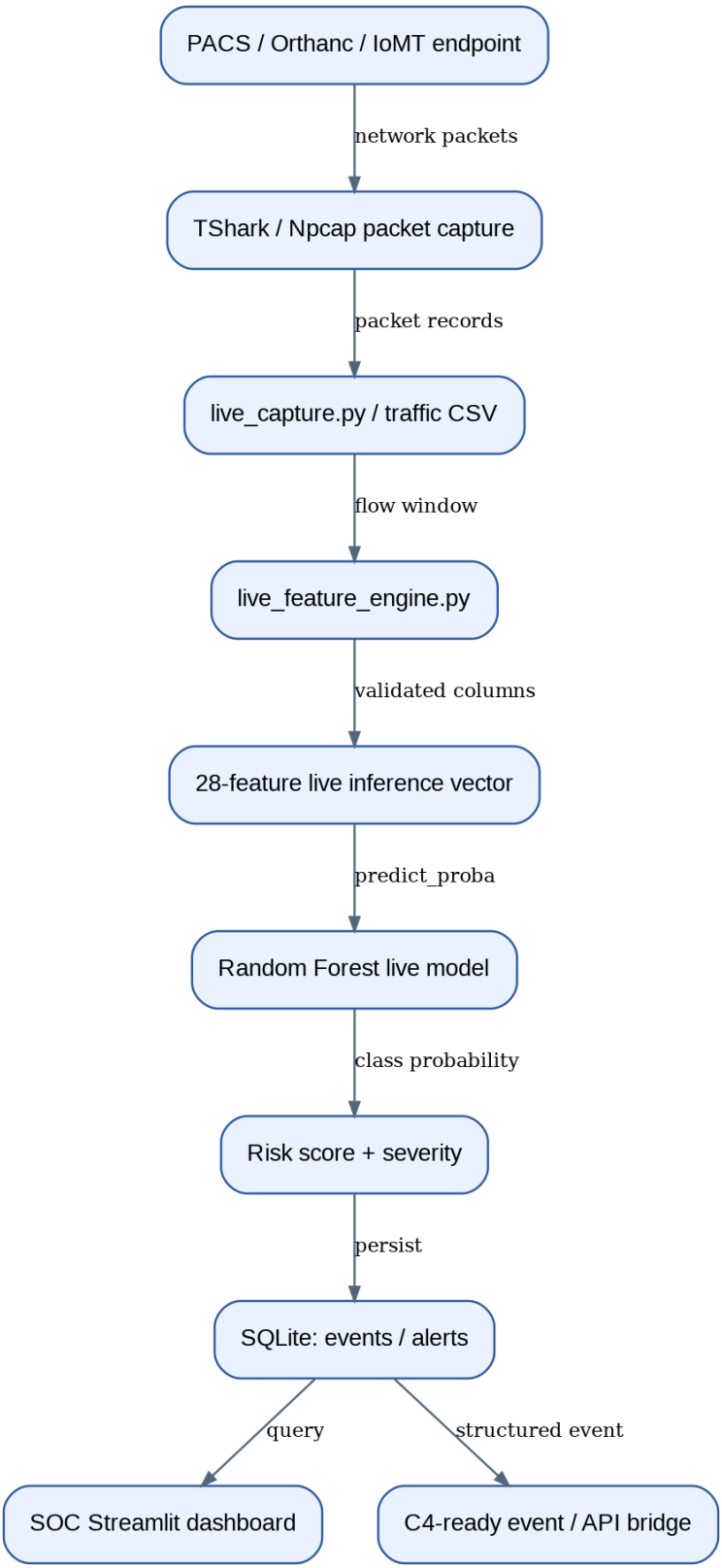


Figure 4.1: Detailed architecture of Component 01.

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4.3 Design alternatives and decisions

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4.4 Detailed design

The DICOM study-integrity path is logically distinct from network anomaly classification. At the trusted source, the system records an original file digest using SHA-256 and associates it with a study identifier. When a received file is provided, the same hash algorithm processes its bytes; exact digest equality indicates that the compared file contents are identical under the cryptographic hash assumption. A mismatch warrants investigation but does not identify the cause of the modification.

The doctor-facing interface should avoid presenting anomaly probability as a clinical diagnosis. A simple result such as Integrity Verified, Integrity Mismatch, or Verification Pending is more appropriate. Clinical personnel should be informed that a matching hash validates byte-level consistency relative to the recorded reference, not medical correctness or origin authentication by itself.

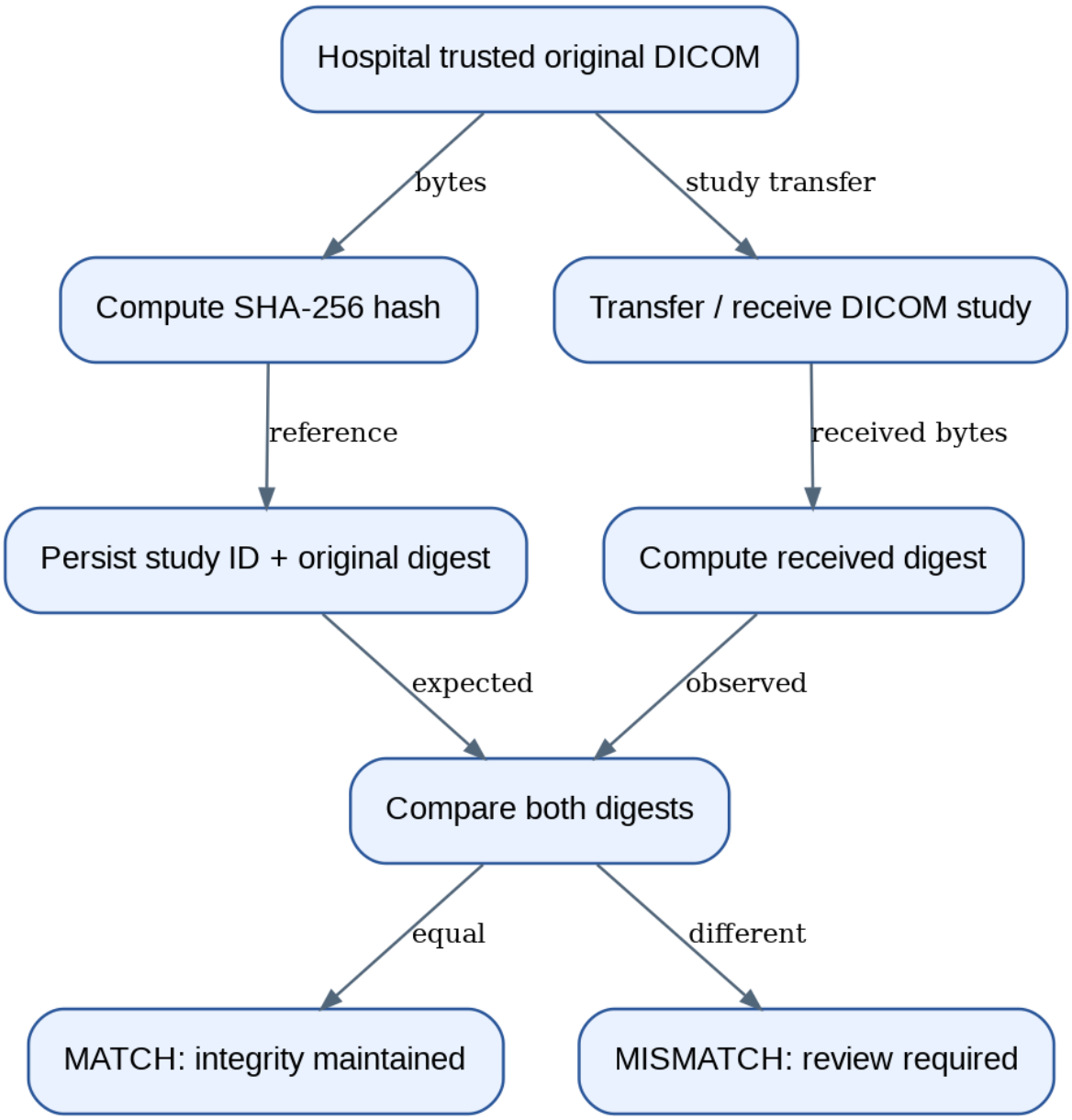


Figure 4.2: DICOM file-integrity validation using SHA-256.

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4.5 Standards and professional practice

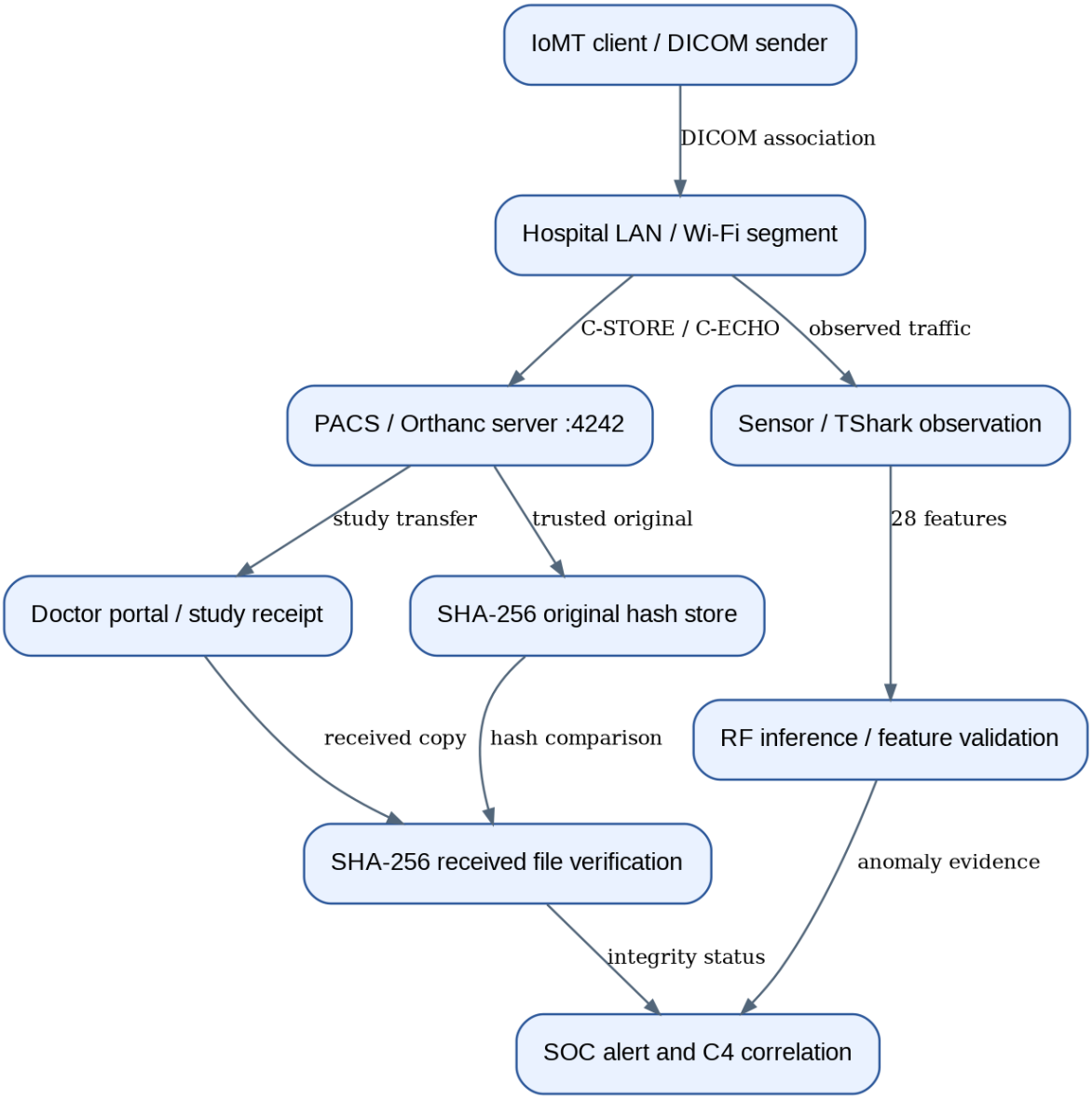


Figure 4.3: Security trust boundaries and medical imaging workflow.

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5 IMPLEMENTATION INTEGRATION AND TESTING

5.1 Development environment and technology stack

The implementation is organised around Python scripts including live_capture.py, live_feature_engine.py, live_predict.py, live_monitor.py, live_soc_bridge.py, soc_app.py and doctor_app.py. The reported technology stack includes pandas, scikit-learn Random Forest, Streamlit, SQLite, Orthanc and Wireshark/TShark or compatible packet capture. The exact package versions should be captured from the delivered environment.

The monitor executes a bounded capture window, validates fresh records, extracts the expected 28 live features, invokes the saved model and writes an inference result. Earlier tests exposed a TCP flag extraction defect in which Boolean flag fields were all reported as zero. After correction, the same 288-packet capture produced nonzero SYN, ACK, FIN, RST and PSH statistics and the classifier reported ABNORMAL with 75.00% abnormal-class probability. This is a pipeline regression check, not proof that the traffic was malicious.

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5.2 Component implementation

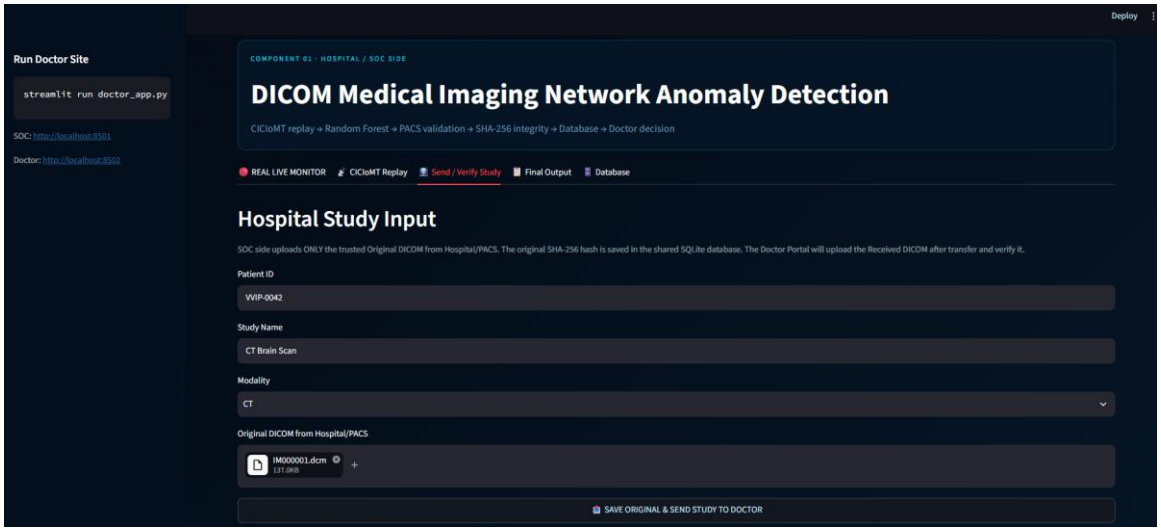


Figure: Original DICOM study input and registration.

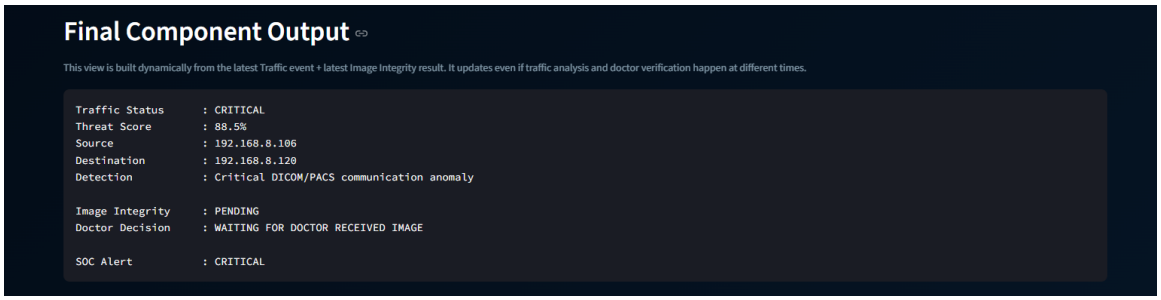


Figure: Combined network and image-integrity decision panel.

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5.3 Integration with the other components

Component 01 contributes network-layer evidence to the cross-layer correlation service. The event representation should include a stable event identifier, event timestamp with timezone, component identifier C1, layer network, event type, source and destination identifiers, destination port, observed DICOM flag, DICOM packet count, total packet count, model version, status, risk score and severity. The consumer should treat missing values explicitly rather than silently substituting zeros for unknown measurements.

Integration should be evaluated separately from local inference: successful SQLite persistence does not by itself demonstrate that C4 received the event. Evidence for a completed API integration consists of an HTTP success response, matching event ID in the receiving database and a displayed correlated record. The supplied screenshots show C4-side records, but a formal end-to-end acceptance test should additionally record request/response logs and schema version.

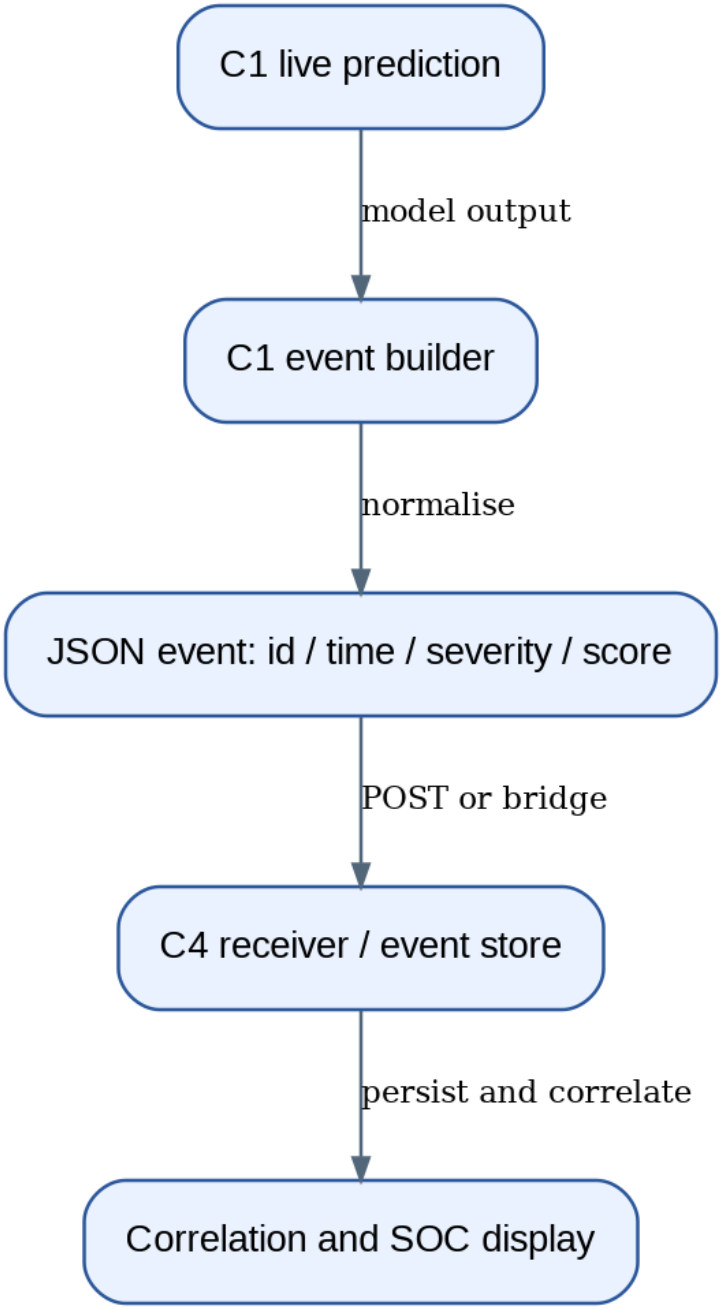


Figure 5.1: Structured event flow between C1 and C4.

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5.4 Verification and software testing

Table 5.1: Component 01 verification scenarios

Test	**Evidence / purpose**
DICOM C-ECHO	Verify association and connectivity to Orthanc
DICOM C-STORE	Verify medical image transfer to Orthanc
Live traffic capture	Capture network traffic and derive 28 features
Random Forest inference	Generate NORMAL/ABNORMAL classification
SOC and C4 forwarding	Record alert and structured integration event

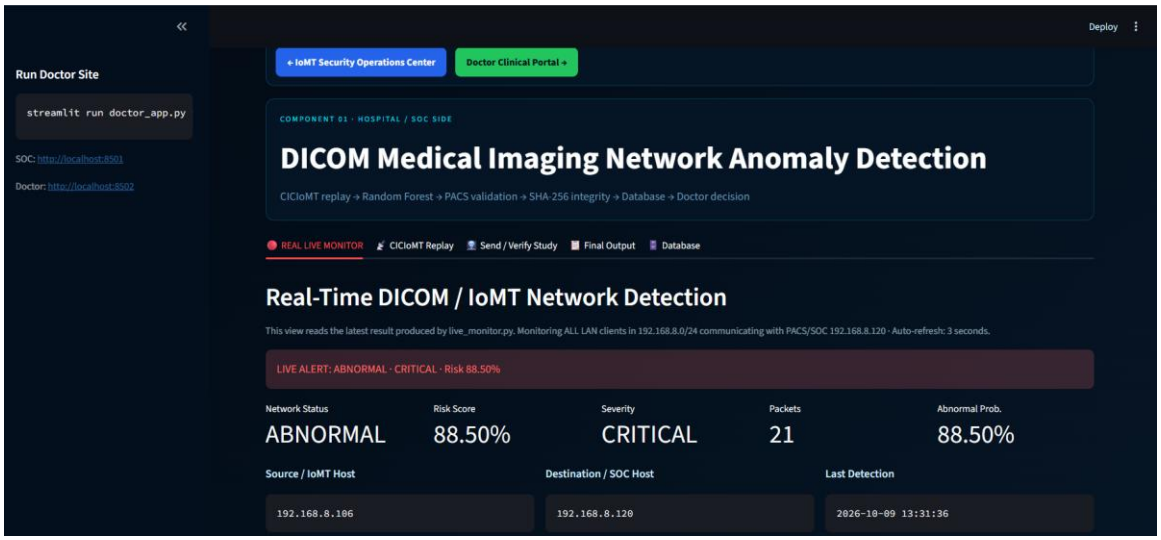


Figure: SOC Analyst Portal live monitoring and latest anomaly.

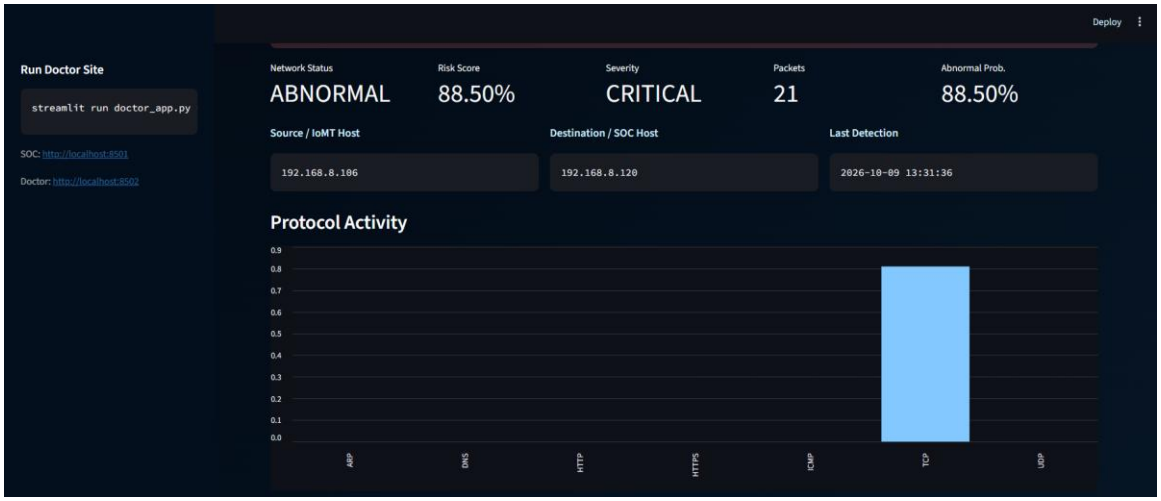


Figure: Network protocol activity visualisation.

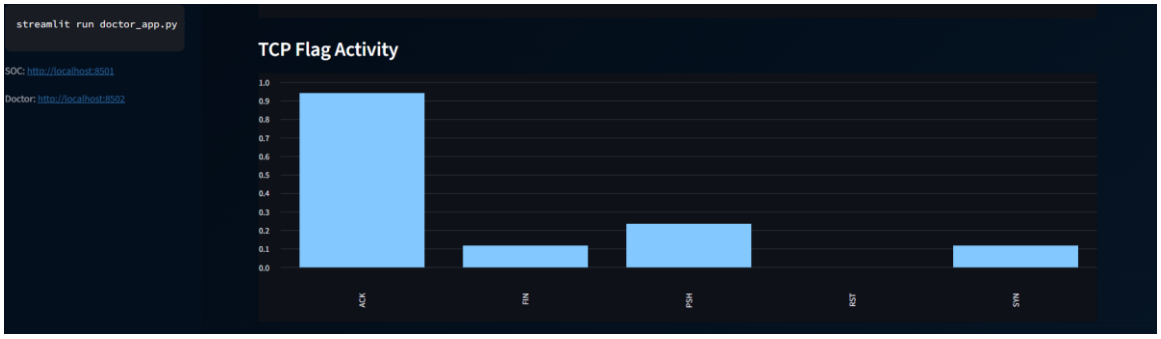
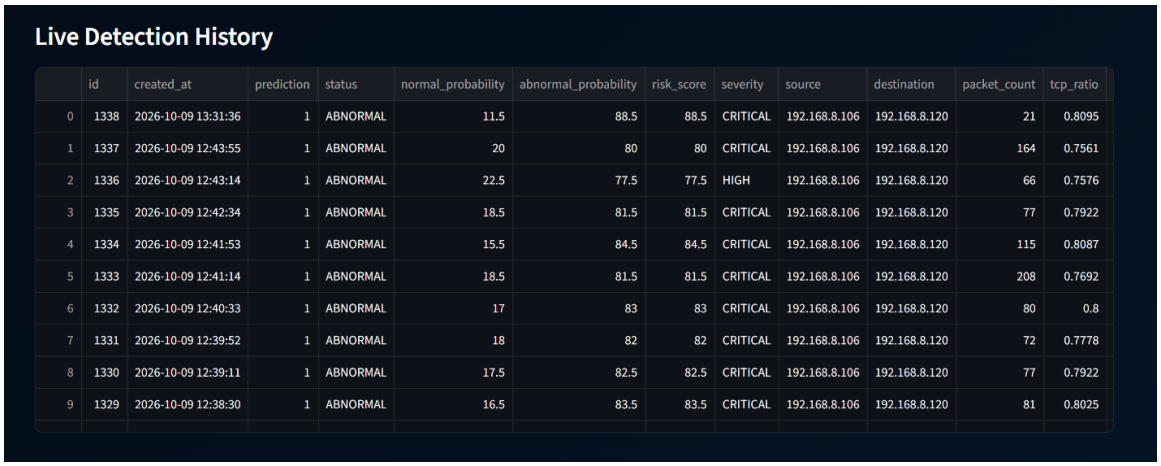


Figure: TCP flag activity extracted from captured traffic.



Live Detection History

	id	created_at	prediction	status	normal_probability	abnormal_probability	risk_score	severity	source	destination	packet_count	tcp_ratio
0	1338	2026-10-09 13:31:36	1	ABNORMAL	11.5	88.5	88.5	CRITICAL	192.168.8.106	192.168.8.120	21	0.8095
1	1337	2026-10-09 12:43:55	1	ABNORMAL	20	80	80	CRITICAL	192.168.8.106	192.168.8.120	164	0.7561
2	1336	2026-10-09 12:43:14	1	ABNORMAL	22.5	77.5	77.5	HIGH	192.168.8.106	192.168.8.120	66	0.7576
3	1335	2026-10-09 12:42:34	1	ABNORMAL	18.5	81.5	81.5	CRITICAL	192.168.8.106	192.168.8.120	77	0.7922
4	1334	2026-10-09 12:41:53	1	ABNORMAL	15.5	84.5	84.5	CRITICAL	192.168.8.106	192.168.8.120	115	0.8087
5	1333	2026-10-09 12:41:14	1	ABNORMAL	18.5	81.5	81.5	CRITICAL	192.168.8.106	192.168.8.120	208	0.7692
6	1332	2026-10-09 12:40:33	1	ABNORMAL	17	83	83	CRITICAL	192.168.8.106	192.168.8.120	80	0.8
7	1331	2026-10-09 12:39:52	1	ABNORMAL	18	82	82	CRITICAL	192.168.8.106	192.168.8.120	72	0.7778
8	1330	2026-10-09 12:39:11	1	ABNORMAL	17.5	82.5	82.5	CRITICAL	192.168.8.106	192.168.8.120	77	0.7922
9	1329	2026-10-09 12:38:30	1	ABNORMAL	16.5	83.5	83.5	CRITICAL	192.168.8.106	192.168.8.120	81	0.8025

Figure: Persisted live detection history.

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5.5 Deployment and operation

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6 RESULTS AND VALIDATION

6.1 Research results

The available operational evidence demonstrates successful execution of live packet capture, construction of 28 network features, model inference and persistence of the classification outcome. The screenshots document a recent 21-packet capture classified ABNORMAL with an abnormal-class score of 88.5%, displayed as CRITICAL by the application severity policy. This is a single operational observation rather than a dataset-wide performance estimate.

Exact model training accuracy, precision, recall, F1, class support, cross-validation variance and feature importances cannot be reconstructed from the dashboard screenshots. These metrics should be inserted only after inspecting the final notebook, dataset and model. A formal validation appendix must preserve the exported confusion matrix, split methodology, software versions and random seed.

A documented two-host test used a SOC/PACS host and a client host connected over Wi-Fi. The IP addresses changed with the local network and should not be treated as fixed deployment parameters. Successful C-ECHO and C-STORE exchanges confirmed basic PACS connectivity. A later live capture contained 308 packets; the extractor produced 28 features and reported a 74.50% abnormal-class probability, ABNORMAL classification and HIGH configured severity.

These numbers are individual run observations, not aggregate model accuracy. No trustworthy held-out confusion matrix, sensitivity, specificity, class balance or calibration report is included in the supplied thesis template. Therefore such values are deliberately not fabricated. A future validated results table should be populated directly from notebook outputs, with test-set sizes and positive-class definitions clearly stated.

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6.2 Product and technical results

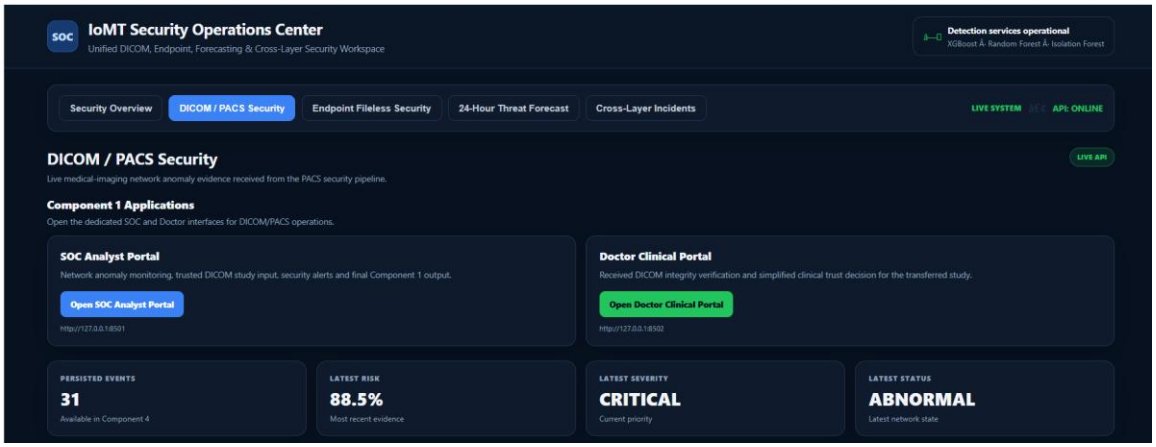


Figure: Integrated MediRecon SOC workspace showing Component 1 overview.

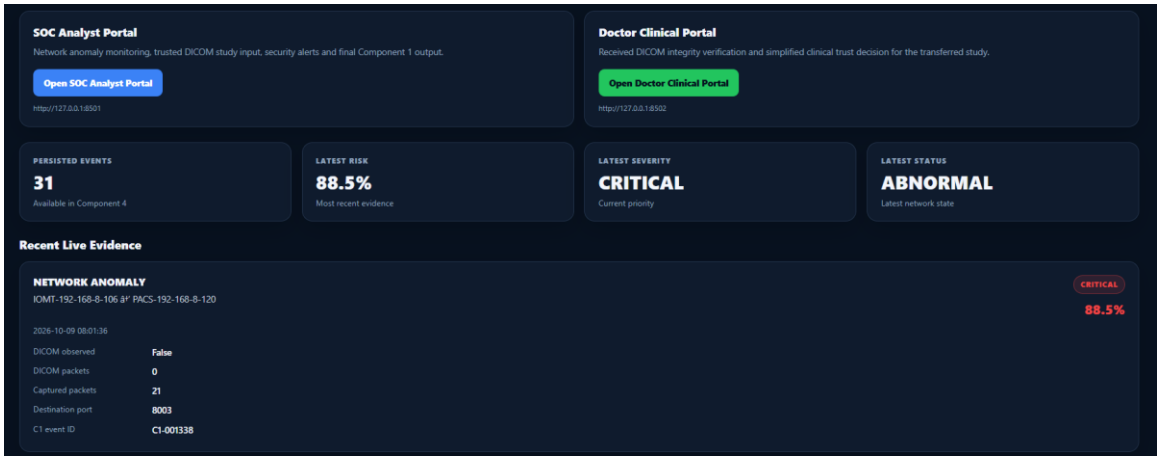
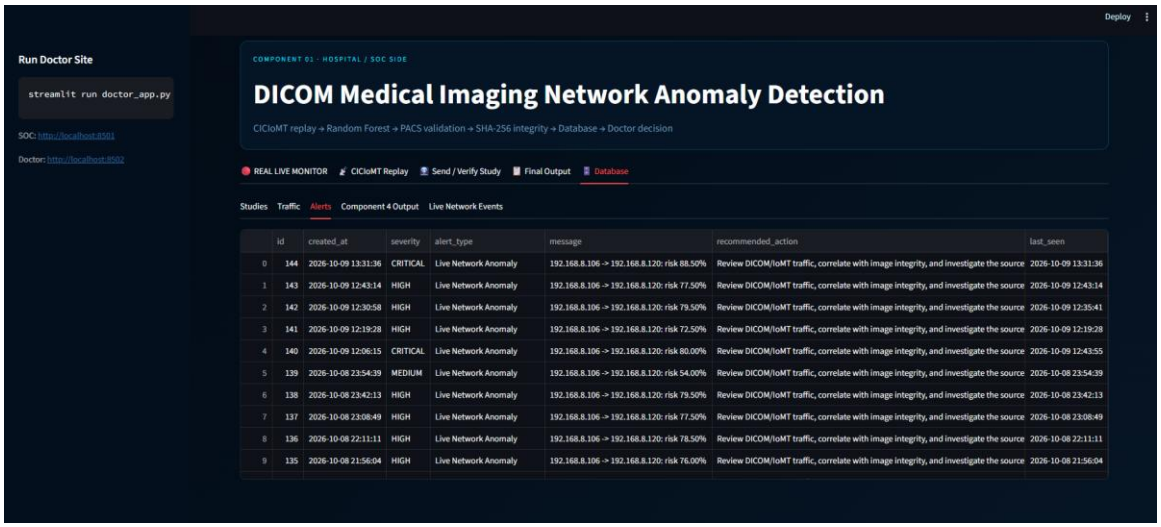
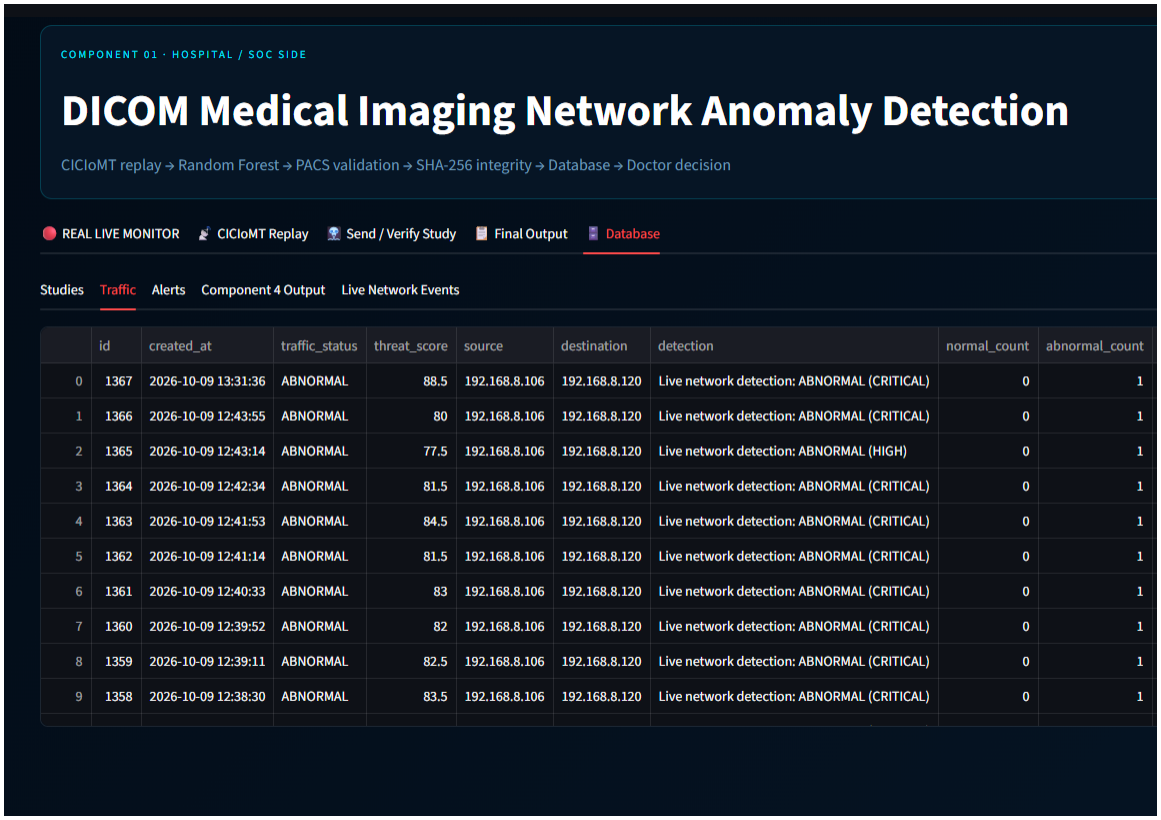


Figure: C1 event summary and recent live evidence.



	id	created_at	severity	alert_type	message	recommended_action	last_seen
0	144	2026-10-09 13:31:36	CRITICAL	Live Network Anomaly	192.168.8.106 -> 192.168.8.120: risk 88.50%	Review DICOM/IoMT traffic, correlate with image integrity, and investigate the source	2026-10-09 13:31:36
1	143	2026-10-09 12:43:14	HIGH	Live Network Anomaly	192.168.8.106 -> 192.168.8.120: risk 77.50%	Review DICOM/IoMT traffic, correlate with image integrity, and investigate the source	2026-10-09 12:43:14
2	142	2026-10-09 12:30:58	HIGH	Live Network Anomaly	192.168.8.106 -> 192.168.8.120: risk 79.50%	Review DICOM/IoMT traffic, correlate with image integrity, and investigate the source	2026-10-09 12:35:41
3	141	2026-10-09 12:19:28	HIGH	Live Network Anomaly	192.168.8.106 -> 192.168.8.120: risk 72.50%	Review DICOM/IoMT traffic, correlate with image integrity, and investigate the source	2026-10-09 12:19:28
4	140	2026-10-09 12:06:15	CRITICAL	Live Network Anomaly	192.168.8.106 -> 192.168.8.120: risk 80.00%	Review DICOM/IoMT traffic, correlate with image integrity, and investigate the source	2026-10-09 12:43:55
5	139	2026-10-08 23:54:39	MEDIUM	Live Network Anomaly	192.168.8.106 -> 192.168.8.120: risk 54.00%	Review DICOM/IoMT traffic, correlate with image integrity, and investigate the source	2026-10-08 23:54:39
6	138	2026-10-08 23:42:13	HIGH	Live Network Anomaly	192.168.8.106 -> 192.168.8.120: risk 79.50%	Review DICOM/IoMT traffic, correlate with image integrity, and investigate the source	2026-10-08 23:42:13
7	137	2026-10-08 23:08:49	HIGH	Live Network Anomaly	192.168.8.106 -> 192.168.8.120: risk 77.50%	Review DICOM/IoMT traffic, correlate with image integrity, and investigate the source	2026-10-08 23:08:49
8	136	2026-10-08 22:11:11	HIGH	Live Network Anomaly	192.168.8.106 -> 192.168.8.120: risk 78.50%	Review DICOM/IoMT traffic, correlate with image integrity, and investigate the source	2026-10-08 22:11:11
9	135	2026-10-08 21:56:04	HIGH	Live Network Anomaly	192.168.8.106 -> 192.168.8.120: risk 76.00%	Review DICOM/IoMT traffic, correlate with image integrity, and investigate the source	2026-10-08 21:56:04

Figure: Persisted security alerts with analyst actions.



	id	created_at	traffic_status	threat_score	source	destination	detection	normal_count	abnormal_count
0	1367	2026-10-09 13:31:36	ABNORMAL	88.5	192.168.8.106	192.168.8.120	Live network detection: ABNORMAL (CRITICAL)	0	1
1	1366	2026-10-09 12:43:55	ABNORMAL	80	192.168.8.106	192.168.8.120	Live network detection: ABNORMAL (CRITICAL)	0	1
2	1365	2026-10-09 12:43:14	ABNORMAL	77.5	192.168.8.106	192.168.8.120	Live network detection: ABNORMAL (HIGH)	0	1
3	1364	2026-10-09 12:42:34	ABNORMAL	81.5	192.168.8.106	192.168.8.120	Live network detection: ABNORMAL (CRITICAL)	0	1
4	1363	2026-10-09 12:41:53	ABNORMAL	84.5	192.168.8.106	192.168.8.120	Live network detection: ABNORMAL (CRITICAL)	0	1
5	1362	2026-10-09 12:41:14	ABNORMAL	81.5	192.168.8.106	192.168.8.120	Live network detection: ABNORMAL (CRITICAL)	0	1
6	1361	2026-10-09 12:40:33	ABNORMAL	83	192.168.8.106	192.168.8.120	Live network detection: ABNORMAL (CRITICAL)	0	1
7	1360	2026-10-09 12:39:52	ABNORMAL	82	192.168.8.106	192.168.8.120	Live network detection: ABNORMAL (CRITICAL)	0	1
8	1359	2026-10-09 12:39:11	ABNORMAL	82.5	192.168.8.106	192.168.8.120	Live network detection: ABNORMAL (CRITICAL)	0	1
9	1358	2026-10-09 12:38:30	ABNORMAL	83.5	192.168.8.106	192.168.8.120	Live network detection: ABNORMAL (CRITICAL)	0	1

Figure: Persisted traffic classification evidence.

A documented two-host test used a SOC/PACS host and a client host connected over Wi-Fi. The IP addresses changed with the local network and should not be treated as fixed deployment parameters. Successful C-ECHO and C-STORE exchanges confirmed basic PACS connectivity. A later live capture contained 308 packets; the extractor produced 28 features and

reported a 74.50% abnormal-class probability, ABNORMAL classification and HIGH configured severity.

These numbers are individual run observations, not aggregate model accuracy. No trustworthy held-out confusion matrix, sensitivity, specificity, class balance or calibration report is included in the supplied thesis template. Therefore such values are deliberately not fabricated. A future validated results table should be populated directly from notebook outputs, with test-set sizes and positive-class definitions clearly stated.

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Verification checklist: preserve training/test split and random seed; report class frequencies; inspect feature ordering; capture fitted hyperparameters; include per-class precision, recall, F1 and confusion matrix; test calibration and false-positive rates; repeat results across multiple controlled captures.

6.3 User or stakeholder validation

A documented two-host test used a SOC/PACS host and a client host connected over Wi-Fi. The IP addresses changed with the local network and should not be treated as fixed deployment parameters. Successful C-ECHO and C-STORE exchanges confirmed basic PACS connectivity. A later live capture contained 308 packets; the extractor produced 28 features and reported a 74.50% abnormal-class probability, ABNORMAL classification and HIGH configured severity.

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6.4 Integration results

The supplied interface captures show persisted C1 traffic records, alert records, live network event history and a Component 4 output view. The visible latest event identifies source 192.168.8.106 and destination 192.168.8.120, and shows destination port 8003 with DICOM observed false and zero DICOM packets. This demonstrates an IoMT/network anomaly classification, not confirmed DICOM packet inspection for that particular window.

Other reported project tests established DICOM C-ECHO and C-STORE operation against an Orthanc instance on port 4242. These DICOM communication tests should be documented as a distinct scenario from the HTTP/API traffic shown in the 8003 event screenshot. The separation is essential for defensible claims about protocol-specific detection.

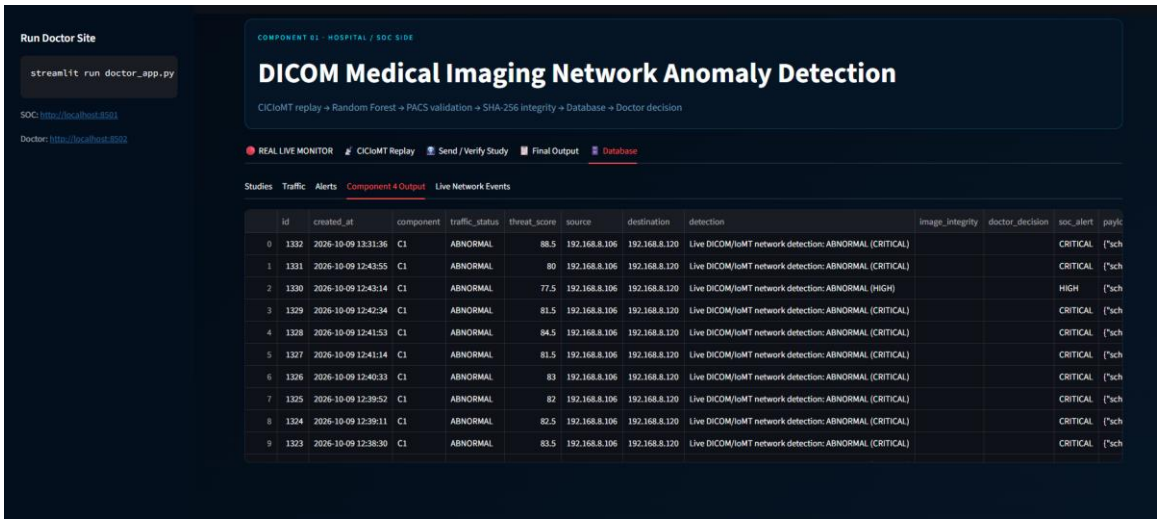


Figure: C4 integration event records.

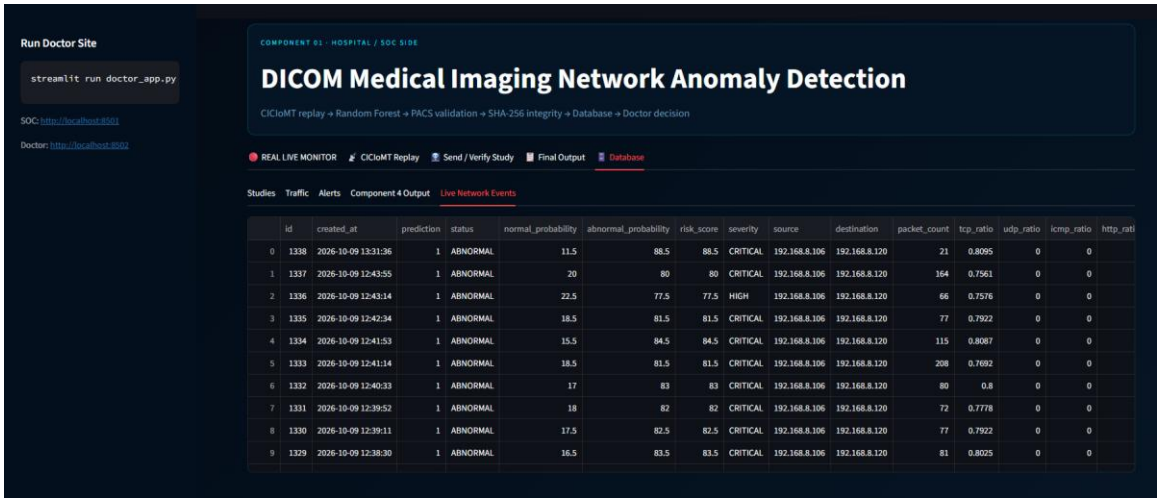


Figure: Live network events and protocol feature evidence.

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6.5 Summary of findings

A documented two-host test used a SOC/PACS host and a client host connected over Wi-Fi. The IP addresses changed with the local network and should not be treated as fixed deployment parameters. Successful C-ECHO and C-STORE exchanges confirmed basic PACS connectivity. A later live capture contained 308 packets; the extractor produced 28 features and reported a 74.50% abnormal-class probability, ABNORMAL classification and HIGH configured severity.

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7 DISCUSSION

7.1 Interpretation of findings

The principal engineering achievement is an observable path from actual network capture through a fixed numerical feature schema to a SOC-facing classification. The component also treats DICOM file integrity as a separate cryptographic problem rather than implying that a traffic classifier can detect image alteration. These distinctions improve technical explainability and make the individual contribution easier to evaluate.

Limitations include dataset-to-deployment shift, imperfect correspondence between CICIoMT benchmark traffic and hospital DICOM workflows, false positives, uncertain probability calibration, limited external validation, and the possibility of attacks that leave packet aggregates or file bytes unchanged. A SHA-256 match proves equality to the trusted reference at the time of comparison; it does not by itself authenticate the party that supplied that reference.

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7.2 Comparison with literature baselines and existing solutions

The principal engineering achievement is an observable path from actual network capture through a fixed numerical feature schema to a SOC-facing classification. The component also treats DICOM file integrity as a separate cryptographic problem rather than implying that a traffic classifier can detect image alteration. These distinctions improve technical explainability and make the individual contribution easier to evaluate.

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7.3 Research contribution

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7.5 Limitations and threats to validity

External validity is limited by the gap between a public IoMT dataset and traffic generated by an actual clinical PACS environment. Controlled two-laptop testing does not reproduce hospital network segmentation, equipment diversity, encrypted communication, maintenance windows or adversarial behaviour. Packet-capture permissions, Wi-Fi visibility and NIC driver behaviour can change the observed packet sample.

Model explainability requires more than reporting the number of features. The final thesis should show feature definitions, data types, feature importance or SHAP explanations if actually computed, representative decision paths and known false-positive patterns. Without these, the supervisor warning about unclear model behaviour remains only partially addressed.

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7.6 Implications

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8 COMMERCIALISATION AND SUSTAINABILITY

8.1 Value proposition and target users

MediRecon targets hospital SOC teams, PACS administrators, healthcare IT security teams and clinical staff requiring an understandable image-integrity status. Its potential value lies in correlating operational network warnings with image integrity and other endpoint evidence, while retaining a clinician-friendly decision surface.

Commercial deployment would require formal procurement review, support commitments, logging governance, patient-data protection, robust access controls, validated incident-response procedures, monitoring for model drift and a security maintenance process. This thesis makes no claim of regulatory approval, clinical certification or production deployment. The proposed direction supports SDG 3 through healthcare safety and SDG 9 through resilient digital infrastructure.

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8.2 Market and competitive position

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8.3 Business or sustainability model

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8.4 Scalability intellectual property and compliance

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8.5 Contribution to the Sustainable Development Goals

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9 CONCLUSION AND FUTURE DIRECTIONS

9.1 Conclusion

Component 01 demonstrates the feasibility of integrating network telemetry classification, PACS communication validation and cryptographic image-integrity checking into a healthcare security workflow. The strongest evidence is the verified end-to-end software operation in a controlled two-host environment, rather than a claim of proven clinical or enterprise-scale efficacy.

The immediate priorities are to archive the final model configuration and training notebook, reproduce held-out metrics, document the feature-schema correspondence, test negative and adversarial DICOM scenarios, calibrate alert thresholds, and conduct a stakeholder usability study. These activities are essential to convert an engineering prototype into a defensible and auditable healthcare security component.

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9.2 Achievement of objectives

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9.3 Individual research and product contributions

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9.4 Future directions

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The immediate priorities are to archive the final model configuration and training notebook, reproduce held-out metrics, document the feature-schema correspondence, test negative and adversarial DICOM scenarios, calibrate alert thresholds, and conduct a stakeholder usability study. These activities are essential to convert an engineering prototype into a defensible and auditable healthcare security component.

For this subsection, the evidence boundary is explicit: the provided thesis template contains headings and guidance rather than primary experimental artefacts. Consequently, precise implementation assertions must be checked against the final source repository, notebook, model metadata and logged outputs. Any quantitative claim should identify the observation window, input source, model version and method of calculation.

In a hospital context, a useful security decision requires an auditable distinction among measurement, classification and response. Packet-derived features describe observed network behaviour; a classifier estimates a learned class; severity and incident response are operational policies. A correct deployment should retain all three as separately inspectable fields and should support analyst review before high-impact containment.

REFERENCES

Editorial verification note: every cited publication and standard must be checked against its original bibliographic record before submission; the draft references must not be presented as independently verified literature searches.

[1] CICIoMT2024 dataset authors, “CICIoMT2024: A benchmark dataset for multi-protocol security assessment in IoMT,” Internet of Things, 2024. doi:10.1016/j.iot.2024.101351. https://www.sciencedirect.com/science/article/pii/S2542660524002920

[2] National Electrical Manufacturers Association, “DICOM PS3.1: Introduction and Overview.” https://dicom.nema.org/medical/dicom/current/output/html/part01.html

[3] National Electrical Manufacturers Association, “DICOM PS3.15: Security and System Management Profiles.” https://dicom.nema.org/medical/dicom/current/output/html/part15.html

[4] L. Breiman, “Random Forests,” Machine Learning, vol. 45, pp. 5–32, 2001. doi:10.1023/A:1010933404324.

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GLOSSARY

Abnormal-class probability — fraction of trained forest trees voting for the abnormal class, subject to the implementation of the classifier; it is not automatically a calibrated real-world attack likelihood.

Feature schema — ordered set of numerical inputs expected by a fitted classifier.

Integrity reference — trusted original DICOM file or digest used as the baseline for a comparison.

Live window — bounded period during which packets are collected and aggregated into one model input.

Domain shift — difference between development data and traffic observed in the intended deployment environment.

APPENDIX

Training notebook: 01_DICOM_RF_Model_Training.ipynb.

Dataset: CICIoMT_DICOM_600K_Clean.csv [verify checksum, row count, labels and licensing].

Live scripts: live_capture.py, live_feature_engine.py, live_predict.py, live_monitor.py.

Observed capture: 288 packets, corrected TCP flag extraction; model result ABNORMAL, 75.00% abnormal-class probability.

Observed capture: 308 packets, 28 generated features; model result ABNORMAL, 74.50% abnormal-class probability.

Evidence still required: final class counts, complete feature list, model hyperparameters, held-out performance and screenshots.

APPENDIX B — ACADEMIC INTEGRITY AND AI DISCLOSURE

Generative AI was used to help organise and draft thesis prose from the supplied template and project descriptions. The author must verify each technical statement, replace unverified placeholders, independently review and cite literature, and disclose representative workflows in the format required by the institution. AI-generated prose must not be represented as independent experimental evidence.

APPENDIX C — TEST CASE MATRIX

Normal network / matching trusted DICOM: expected low network concern and verified integrity.

Abnormal network / matching trusted DICOM: expected network alert, integrity remains verified.

Normal network / modified DICOM: expected integrity failure regardless of network classification.

Abnormal network / modified DICOM: expected combined escalation subject to configured policy.

No fresh packets: skip inference and report unavailable window rather than reusing stale capture.

Incorrect feature schema: reject inference and record a diagnostic error.

APPENDIX D — FEATURE CONTRACT AND MODEL AUDIT

The deployed Component 01 pipeline uses the following 28 live feature names. Descriptions explain their analytical roles; exact aggregation formulas and units should be taken from the deployed feature-engine code rather than inferred from column names.

D.1 Duration

Duration represents the elapsed time covered by the observed packet sequence or flow. It provides temporal context for the capture and should be interpreted alongside packet volume rather than as an attack indicator.

D.2 Rate

Rate describes traffic intensity over the measurement interval. A sudden increase can indicate a burst of connections or transfers, but scheduled medical-image uploads can also produce high rates.

D.3 fin_flag_number

This field reflects TCP FIN-flag activity, associated with orderly connection closure. An unusual concentration of FIN packets may indicate short-lived sessions or connection churn.

D.4 syn_flag_number

SYN activity captures attempts to initiate TCP connections. Elevated SYN presence without a corresponding progression to established sessions can warrant investigation.

D.5 rst_flag_number

TCP RST flags indicate abrupt connection resets. Frequent resets may result from refused services, unstable communication, or scanning behaviour.

D.6 psh_flag_number

The PSH flag is commonly associated with TCP segments carrying data for prompt delivery. In the DICOM testbed, PSH observations help distinguish payload-bearing traffic from connection-control packets.

D.7 ack_flag_number

ACK-flag activity reflects acknowledgement behaviour within TCP sessions. Its relationship to SYN and data-bearing packets provides context for connection establishment and sustained communication.

D.8 ece_flag_number

The ECE flag relates to TCP Explicit Congestion Notification. Its occurrence may reflect congestion signalling rather than malicious activity, so interpretation depends on network conditions.

D.9 cwr_flag_number

CWR indicates a sender response to congestion notification. This feature contributes congestion-related context and is not a direct indicator of fileless malware.

D.10 ack_count

ACK count measures acknowledgement events in the observed traffic. It is useful for characterising ongoing TCP exchanges and comparing session behaviour across capture windows.

D.11 syn_count

SYN count represents connection-initiation events. Repeated attempts over a short interval can help highlight unusual access patterns, particularly when paired with destination and reset information.

D.12 fin_count

FIN count records orderly session-termination events. The count can help distinguish many short transactions from fewer long-lived sessions.

D.13 rst_count

RST count records forced or unexpected TCP connection termination. It is relevant when assessing failed connections, rejected requests, and unstable endpoints.

D.14 HTTP

HTTP activity represents unencrypted web-protocol traffic detected in the window. In this testbed it can include local application access and should be separated from DICOM protocol traffic.

D.15 HTTPS

HTTPS activity identifies encrypted web communication. Because application payloads are not directly visible, the feature primarily provides protocol-distribution context.

D.16 DNS

DNS activity reflects name-resolution traffic. Bursts of DNS queries may accompany routine service discovery or suspicious communication and require supporting evidence.

D.17 TCP

TCP represents transport-layer traffic using reliable, connection-oriented delivery. DICOM C-ECHO and C-STORE exchanges with Orthanc operate over TCP, making correct TCP mapping important to the live feature pipeline.

D.18 UDP

UDP represents connectionless transport traffic, such as some discovery and name-resolution exchanges. Unlike DICOM over TCP, UDP activity mainly supplies background network context in the described testbed.

D.19 ARP

ARP records local address-resolution activity linking IPv4 addresses to link-layer addresses. Its presence can help characterise local network communication but does not establish a DICOM transaction.

D.20 ICMP

ICMP captures network control and diagnostic messages, including some reachability tests. Unusual ICMP volumes may be worth investigating separately from clinical imaging traffic.

D.21 IGMP

IGMP relates to IPv4 multicast group management. It is generally background network-control activity and should not be confused with DICOM associations.

D.22 Min

Min is the smallest observed value of the packet-size statistic used by the feature engine. It helps characterise the lower end of traffic-size distribution.

D.23 Max

Max is the largest observed packet-size value. Large packets may occur during legitimate DICOM transfers, so the maximum must be considered with the full size distribution.

D.24 AVG

AVG is the mean of the packet-size observations in the capture window. It summarises typical packet size while smoothing variation among individual packets.

D.25 Std

Std measures dispersion around the mean packet size. Greater variation may reflect a mixture of control messages and data transfers.

D.26 Tot size

Tot size aggregates packet sizes across the observed window. It provides a volume measure useful for comparing light control traffic with larger image-transfer sessions.

D.27 Number

Number denotes the count of packet observations included in the feature window. It is also relevant to the insufficient-packet guard used before inference.

D.28 Variance

Variance measures spread in packet-size observations and is related to, but distinct from, standard deviation. It captures changes in the distribution of packet sizes.

APPENDIX E — REPRODUCIBLE ACCEPTANCE TEST PROTOCOLS

The following concise protocols describe the expected checks. A successful test should be supported by its corresponding execution output or persisted event; the protocol itself is not evidence of a completed test.

E.1 Connectivity and interface selection

Record the client and Orthanc server IP addresses, verify reachability, and select the active capture interface. This establishes that the monitor observes the intended testbed traffic.

E.2 HTTP traffic control

Generate bounded requests to a local HTTP service and inspect the captured protocol mix. This checks that ordinary web traffic is distinguished from DICOM-related activity.

E.3 DICOM C-ECHO

Send a DICOM C-ECHO request to the Orthanc listener on port 4242. A successful response demonstrates DICOM association and basic connectivity, not image transfer.

E.4 DICOM C-STORE

Transmit the sample DICOM object to Orthanc and confirm it appears in the PACS study inventory. This validates the image-storage path and provides genuine DICOM traffic for capture.

E.5 Capture guard

Provide an empty or undersized packet window and verify that inference is skipped. This prevents an apparently confident classification based on insufficient evidence.

E.6 Feature schema

Check that the live feature vector contains all 28 expected columns in the estimator's required order. Missing or misordered values must not silently reach the classifier.

E.7 Prediction persistence

Compare the classifier output with the corresponding stored alert or event. The recorded status, score, timestamp, and packet context should refer to the same capture.

E.8 Alert deduplication

Repeat a comparable traffic event and inspect whether duplicate notifications are suppressed according to the configured policy, without losing meaningful event history.

E.9 Integrity match

Calculate SHA-256 for an original test DICOM object and an unchanged copy. Equal digests establish byte-for-byte agreement for the compared files.

E.10 Integrity mismatch

Modify a test copy and compare its SHA-256 digest with the trusted original. A mismatch should be shown as an integrity concern independently of network classification.

E.11 C4 integration

Submit a C1 event in the agreed JSON schema and inspect its appearance in the cross-layer correlation system. Event identity and severity should remain consistent.

E.12 Doctor workflow

Inspect the doctor-facing view for the available study, network status, and integrity decision. The interface should communicate results without requiring SOC-level technical knowledge.

E.13 Dashboard refresh

Observe successive dashboard updates while monitoring is active. Existing events should remain visible and refreshes should not create duplicate records.

E.14 Time consistency

Compare one event timestamp across the live monitor, database, and integrated dashboard. Any timezone conversion should preserve the event's actual sequence.

E.15 Security and privacy

Use non-identifying test DICOM data and restrict access to stored study metadata. Report only controls that were actually configured and exercised.

APPENDIX F — RESEARCH EVIDENCE REGISTER

The following entries identify the purpose of the supplied interface screenshots. Each image documents the visible state of the system at capture time; model accuracy and backend behaviour require separate experimental evidence.

F.1 Integrated MediRecon SOC workspace

Shows the overall Component 01 status within the combined MediRecon security workspace.

F.2 C1 event summary

Presents the latest network-event context and summary indicators available to an analyst.

F.3 SOC Analyst Portal live monitoring

Displays the latest classification, severity, and packet-level context in the monitoring view.

F.4 Network protocol activity

Visualises the distribution of protocols observed in the selected live capture.

F.5 TCP flag activity

Shows TCP control-flag observations used to interpret the captured connection behaviour.

F.6 Live detection history

Presents previously recorded classifications for reviewing changes across monitoring windows.

F.7 Original DICOM study registration

Illustrates the test-study input and registration workflow in the clinical-facing interface.

F.8 Network and image-integrity panel

Displays network assessment and DICOM integrity as separate but related security findings.

F.9 Security alerts and analyst actions

Shows the alert list and analyst-facing controls for reviewing recorded incidents.

F.10 Traffic classification records

Illustrates how captured network observations are retained with classification outcomes.

F.11 C4 integration records

Displays Component 01 event information within the cross-layer integration view.

F.12 Live network events and features

Shows the recorded live network activity and selected protocol-feature values.